ASAN MEMORIAL COLLEGE OF ARTS & SCIENCE

EN'DECORE

Virtual Try Feature for E-Commerce Using Augmented Reality

By

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Under the guidance of

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System Architecture

En'Decore follows a 3-tier client-side architecture with serverless Firebase backend and third-party AR integration.

Presentation Tier

HTML5 Pages, CSS3 Design System, Google Fonts, Font Awesome Icons, Unsplash CDN Images

Application Logic

Vanilla JavaScript, Firebase SDK v9.22, AR.js, DOM Rendering, LocalStorage Fallback

Data & Services

Firebase Auth, Cloud Firestore, Vectary 3D/AR, Firebase Storage

26+

Project Files

75+

Image Assets

0

Servers Required

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Hardware & Software Requirements

Hardware

- Intel Core i3 (8th Gen) or AMD Ryzen 3
- 4 GB RAM (8 GB recommended)
- 500 MB + 2 GB free disk space
- 1366 × 768 resolution display
- ARCore/ARKit compatible smartphone

Software

- HTML5/CSS3/JavaScript
- Firebase SDK v9.22.0
- Unity & C# for AR
- Vectary 3D Cloud
- Chrome 90+ / Edge 90+

Core Functional Modules

The application is organised into 7 core modules, each handling distinct feature areas of the e-commerce platform.

User Authentication

Firebase Auth with email/password, session management, guest cart merge

Product Catalogue

13 furniture products across 3 categories with filtering and sorting

Cart & Orders

Add/remove items, Firestore sync, LocalStorage fallback, checkout with GST

Augmented Reality

Vectary 3D viewer integration, no app install required

Blog/Journal

6 design-focused articles on AR, vintage furniture, farmhouse design

Contact & FAQ

Multi-field form with Firestore storage, accordion-style FAQ with 4 categories

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Product Catalogue

1

Armchairs

Lounger, Patricia, Futon styles

2

Sofas

Various sizes and configurations

3

Tables

Foyer and coffee tables

Product badges, star ratings, dynamic rendering, and filtering by price and name.

Augmented Reality Experience

Key Features

- AR modal with Vectary 3D viewer integration
- Quick-view AR buttons on product cards
- Detail-page AR with full 3D model interaction
- No app installation required
- WebXR support in Chrome 90+

Transforms furniture shopping by eliminating purchase doubt through virtual placement.

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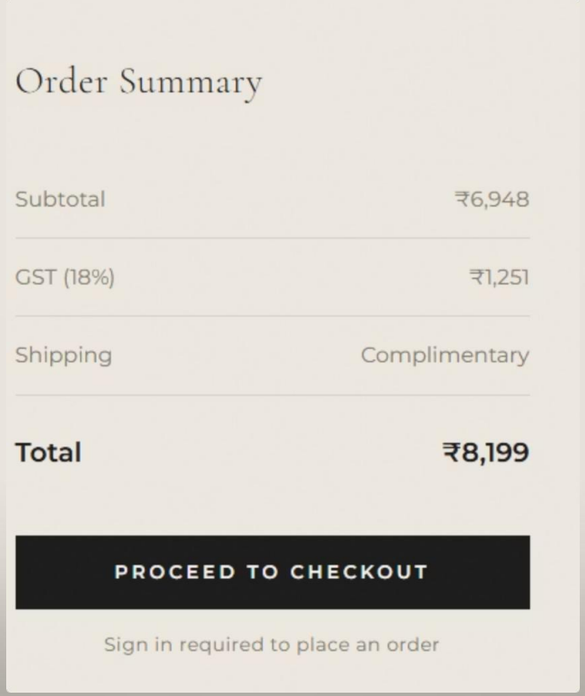

Shopping Cart & Checkout

01

Add to Cart

Real-time Firestore sync when logged in, LocalStorage fallback for guests

02

Update Quantities

Add, remove, or modify item quantities with instant price recalculations

03

Calculate Totals

Subtotal, 18% GST, complimentary shipping applied automatically

04

Place Order

Checkout form with shipping details, order history tracking

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Blog & Journal Content

Six design-focused articles covering AR technology, vintage furniture, farmhouse design, sustainability, furniture arrangement, and home styling.

Augmented Reality in Retail

How AR transforms furniture shopping by eliminating doubt

Vintage Furniture Renaissance

Identifying hidden gems and restoring them to glory

Modern English Farmhouse

Balancing clean lines with soft traditional textures

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Key Features & Benefits

Serverless Architecture

No dedicated server required. Firebase provides complete backend infrastructure as managed service.

AR Integration

Vectary 3D viewer enables web-based augmented reality without app installation.

Responsive Design

Works across desktop, tablet, and mobile devices with consistent experience.

5-Year Warranty

Structural warranty on all furniture pieces for customer confidence.

Thank You

En'Decore

For your time, attention, and valuable feedback. We look forward to your questions and suggestions.

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